Metasploit: added and changed files

Updated: 2 November 2026 (Cairo) · Commits `bf543af` (code) and `59dc2b7` (results)

This page lists the files that were added or changed for the Metasploit integration. Paths are relative to the project root `Cerberus/`.

1. Adapter: how the backend talks to `msfrpcd`

File	Role
`backend/app/ai_engine/pentest/tools/metasploit_adapter.py`	**Replaced in this update.** It was XML-RPC, which does not work with real `msfrpcd`. It now uses MessagePack over HTTP, the protocol `msfrpcd` uses. It keeps the RPC token local, uses bounded timeouts, and refuses console commands with newline or command injection.

2. Scanner module: what can run

File	Role
`backend/app/ai_engine/pentest/modules/tools/metasploit_scanner.py`	**Changed.** Runs only `auxiliary/scanner/portscan/tcp`. Parses IPv4 and IPv6 results (for example `[::1]`). Checks scope. Limits ports to `PENTEST_MAX_PORTS`.

3. Safety and scope files

File	Role
`backend/app/ai_engine/pentest/core/scope_match.py`	**New.** Structured scope matching: host, scheme, path and port. Replaces substring matching.
`backend/app/ai_engine/pentest/core/safety.py`	**Changed.** Refuses any module when `testing_window` is set. Adds the Metasploit and Nmap restrictions.
`backend/app/ai_engine/pentest/core/context.py`	**Changed.** Scope and context checks for the new rules.

4. Settings

File	Change
`backend/app/core/config.py`	`MSF_RPC_TIMEOUT` default is **60 seconds**. The first console is usually quick; it took 0.48 seconds on the test machine.
`backend/.env.example`	Example values for `MSF_ENABLED`, `MSF_RPC_HOST`, `MSF_RPC_PORT`, `MSF_RPC_PASSWORD`, `MSF_RPC_SSL`, `MSF_RPC_TIMEOUT=60`. The password is a placeholder.

File	Change
backend/requirements.txt	Adds msgpack==1.1.2

Your private backend/.env is not changed by this update. If it already sets MSF_RPC_TIMEOUT=10, change it to 60.

5. Tests

File	What it checks
backend/tests/unit/test_pentest_tools.py	Adapter and scanner with a protocol-compatible test double (FakeMsfrpcd). The double does not run Metasploit.
backend/tests/unit/test_pentest_framework.py	Framework engine and scheduler tests (changed)
backend/tests/unit/test_pentest_safety_regressions.py	**New.** Safety regressions: refused modules, refused options, scope and IPv6 cases
backend/tests/unit/test_public_tool_results.py	Tests for the public-capture script (changed)

6. Scripts and documents

File	Role
scripts/live_tests/beginner_tools_lab.py	Beginner lab. Its metasploit mode is **simulated**, not a real daemon.
scripts/live_tests/public_tool_results.py	Refreshes public Nmap and sqlmap captures. It **does not** run Metasploit.
scripts/live_tests/sqlmap_framework_smoke.py	Local smoke test. Its Metasploit endpoint is a **fake** MessagePack service.
docs/real_results.json and docs/REAL_RESULTS.md	Saved results. Metasploit's real loopback runs are listed separately from public results.
scripts/render_real_results.py	Rebuilds the real-results PDF

7. What is not in the repository

- The Metasploit program itself (about 1 GB). It is optional. The beginner lab works without it.
- Temporary loopback test scripts from the real-daemon check. They were removed after the test, so the real-daemon check must be run again by hand (see the steps document).